Acquisition of the blocking effect in L2 Chinese by L1 Indonesian speakers

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Background

This study investigates L1 Indonesian speakers' processing of reflexive *ziji* ('self') in L2 Chinese in relation to the **blocking effect** phenomenon. In Chinese, the reflexive *ziji* can be long-distance (LD) bound by a non-local subject; however, when the local subject is a 1st-person pronoun, LD binding is not allowed (e.g., 'John$_1$ said I$_2$ like self$_{*1/2}$'s shirt'). This is because (i) LD *ziji* is empathic (perspective-sensitive), requiring the speaker or reader to empathize with the non-local subject or ***empathy locus*** and (ii) the 1st-person pronoun 'I' is a **stronger empathy locus** and is priortized over a 3rd-person empathy locus (e.g., Kuno'87). The Indonesian 'half-reflexive' *diri-pro* ('self-pro') can function as a reflexive when locally bound; but when bound non-locally, it functions as a pronoun and does not exhibit blocking effects (e.g., Cole & Hermon'05; Kartono'13). Thus, a crucial question is whether Indonesian-speaking L2 Chinese learners can acquire the empathic features of *ziji*.

To preview the results, we found in three picture-based truth-value judgment experiments that Indonesian speakers can acquire the blocking effect in L2 Chinese and suppressed negative transfer from their L1 Indonesian. However, L2 speakers tended to accept LD binding less often than L1 Chinese speakers, as LD binding requires more cognitive resources.

Methods

Forty-three L2 Chinese learners, in addition to 33 L1 Chinese and 40 L1 Indonesian control participants, participated. Two factors, ANTECEDENT (local vs. non-local) and BLOCKER (blocker vs. non-blocker) were fully crossed in a 2x2 factorial design. See **Fig.1** for an example target set (16 target items and 24 fillers) and how local/non-local binding is created in blocker and non-blocker conditions. In each of the 3 experiments, participants first made a forced choice acceptability judgment and then rated the acceptability of a particular reading using a 7-point Likert scale. The L2 participants also completed a cloze test (L2 proficiency) and a language background survey. Mixed effects logistic and linear regressions were run over forced choice judgments and acceptability ratings, respectively.

Results

We only report results from the forced choice task as acceptability ratings show similar results. In the **L1 Indonesian** experiment (**Fig.2**), we found an ANTECEDENT x BLOCKER interaction ($p < 0.001$): within the local conditions, participants accepted local binding more often in the presence of a blocker due to less ambiguity (i.e., 3rd-person *diri-nya* has two feature-matching 3rd-person antecedents vs. 1st-person *diri-ku* which only has one feature-matching 1st-person antecedent); crucially, within the non-local conditions, there was no blocking effect.

In the **L1 Chinese** experiment (**Fig.3**), in addition to an interaction ($p < 0.001$), we further found a strong blocking effect as Chinese natives strongly rejected non-local binding in the presence of a blocker. **L2 Chinese speakers** (**Fig.4**) showed interpretation patterns of *ziji* similar to L1 Chinese speakers as they also mostly rejected non-local binding when the local subject is 'I'. However, L2 speakers accepted non-local binding in the non-blocker conditions less often than L1 speakers (51% vs. 75%), possibly due to reduced cognitive resources for constructing long-distance dependencies in an L2. Finally, a **correlation analysis** (**Fig.5**) revealed that more proficient L2 speakers exhibited stronger blocking effects ($p < 0.05$). Together, these findings suggest that Indonesian learners of L2 Chinese can acquire the blocking effect associated with *ziji*.

Discussion

We highlight three main findings. First, consistent with prior theoretical discussions of *diri-pro* (e.g., Cole & Hermon 2005; Kartono 2013), our experiments show that *diri-pro* does not show any blocking effect and, unlike *ziji*, cannot function as a LD empathic reflexive. Second, L2 speakers can acquire the empathic features of *ziji*, particularly at higher proficiency levels, in line with predictions of the Feature Reassembly Hypothesis (e.g., Lardiere 2009). Finally, L2 speakers face greater difficulty in establishing long-distance dependencies, which may explain their lower acceptance of non-local binding in the non-blocker conditions.

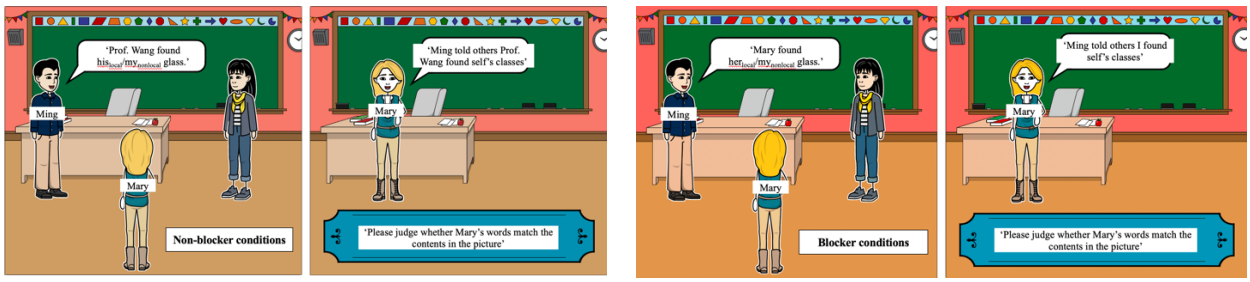
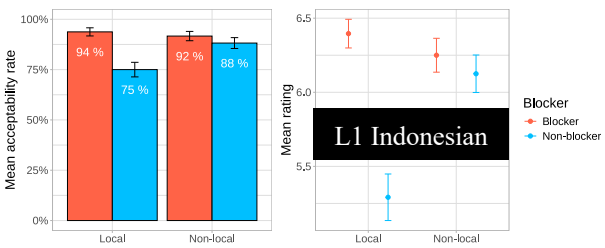


Fig. 1: An example target set with English translations. The original texts were shown in Indonesian and Chinese in respective experiments. The left panel shows two **non-blocker** conditions (i.e., the local subject in Mary's speech is **3rd-person** 'Prof. Wang') with local or non-local antecedents. The right panel shows two **blocker** conditioarns (i.e., the local subject is **1st-person** 'I'). Local or non-local binding readings are contigent upon Ming's speech ('his' indicates local binding; 'my' indicates non-local binding).



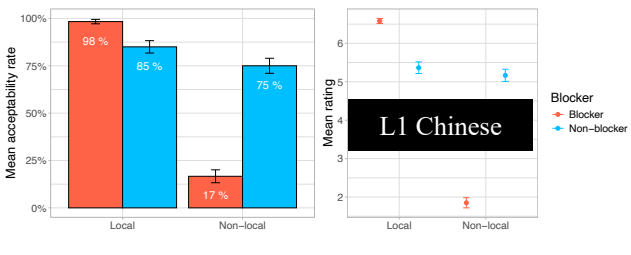
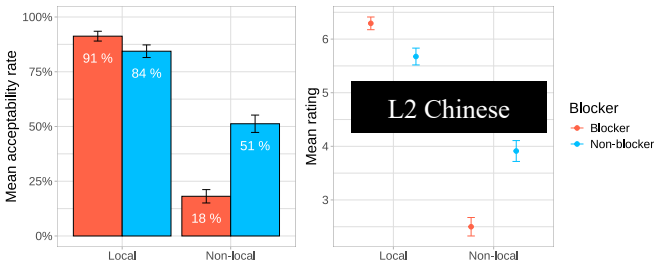


Fig.2 (L1 Indonesian) & **Fig.3** (L1 Chinese): Mean acceptabilities in forced choice and Likert-scale-based rating tasks in L1 Indonesian (*diri*) (left panel) and L1 Chinese (*ziji*) (right panel).



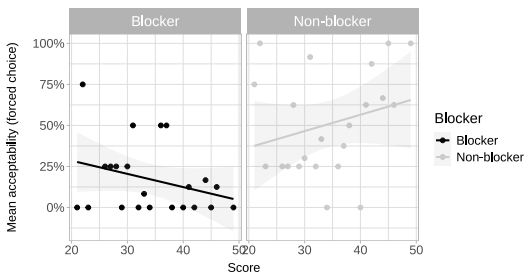


Fig.4 (L2 Chinese) & **Fig.5** (correlation): Mean acceptabilities in L2 Indonesian (*diri*) (left panel) and the correlation between L2 proficiency and mean acceptabilities under the blocker and non-blocker conditions.

References

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